

Model	Summary					Storyworld					Time				
	<32	32-64	64-128	>128	B	<32	32-64	64-128	>128	B	<32	32-64	64-128	>128	B
GPT-4.1	0.80	0.81	0.81	0.82	0.27	0.17	0.27	0.16	0.09	0.00	0.58	0.54	0.54	0.38	0.35
Llama 4 Scout	0.76	0.77	0.74	0.77	0.29	0.10	0.10	0.07	0.02	0.00	0.55	0.60	0.57	0.58	0.24
Gemini 2.0 Flash	0.73	0.75	0.69	0.72	0.27	0.12	0.14	0.09	0.05	0.00	0.55	0.67	0.54	0.63	0.29
DeepSeek V3	0.75	0.80	0.69	0.30	0.24	0.14	0.12	0.08	0.00	0.00	0.50	0.58	0.61	0.45	0.28
Mistral Small 3.1	0.80	0.73	0.62	-	0.24	0.24	0.19	0.11	-	0.01	0.58	0.47	0.59	-	0.26
Gemma 3 27b	0.72	0.66	0.16	-	0.25	0.18	0.25	0.01	-	0.00	0.62	0.54	0.50	-	0.22
Qwen 3 32b	0.78	0.73	0.62	-	0.26	0.25	0.06	0.04	-	0.00	0.56	0.64	0.56	-	0.32

Table 2: LLM performance comparing full novels (T2) to per-chapter results across different volume lengths, with similarity scores for summaries, storyworlds, and times. Values are normalized to the closed interval $[0, 1]$. All reported lengths are in thousands of Gemma 2 tokens. Performance is consistent across models below the 64k–128k length bracket. Compare with random baselines (B) averaged over 1,440 random pairs on a per-model basis.

holds. In fact, differences in model performance between short and long texts increase as the window of interest increases. That is, there is a greater difference in average model performance for <32k and >128k novels when we request analysis of all chapters than when we ask for 25% of chapters. This pattern makes intuitive sense, as the models are forced to consider more text comparatively when examining longer windows of longer texts.

We see limited evidence of memorization from pre-training. Passing only a volume’s title and author to the models (T1) decreases average model performance by roughly a third on the summary task, and to near zero for storyworld; narrative time estimates are equivalent. Similarly, shuffling the chapters of a text (T5) reduces the performance on the summaries and storyworlds for shorter text windows but does not consistently affect time estimates. Truncating the texts to the window of interest (T4) and all treatment combinations with T4 improve performance for truncated windows of interest; this effect is strongest for the shortest windows and longest texts. In contrast, passing individual chapters as user messages (T3) has little consistent impact on performance. Finally, we note that the average model performance when reporting storyworlds falls as the window of interest increases from 25% of the novel to 100%.

5 Discussion and Conclusion

We present the Too Long, Didn’t Model (TLDM) benchmark for long-context understanding. We release initial benchmark scores for seven frontier LLMs released in early 2025. The benchmark includes three narrative understanding tasks: summarization, storyworld reporting, and narrative time estimation. All require models to infer information

over the full text of a novel. It evaluates models on novels of varying lengths (<32k to >128k), assesses the models’ ability to focus on a particular subset of texts, and determines the impact of various experimental treatments (e.g., shuffling chapters). Assessing models via this benchmark therefore provides a comprehensive understanding of their ability to perform complex long-context analysis.

No tested model is perfect at long contexts. Our preliminary model evaluations show that, despite having context windows of up to 10M tokens, models’ performance declines considerably with longer texts (those above 64k tokens), especially for non-summary tasks. The true context of these models is still limited for complex understanding tasks.

Model scale benefits long-context understanding. We find long-context abilities improve linearly with model size. This indicates smaller, open-weight models that can be run on laptops continue to perform worse than do larger commercial models.

Text linearity aids long-context models. We find long-context models are impacted by document order — especially so when focussing on limited narrative windows. This suggests some models develop particularly inelastic mechanisms for tracking linear narratives.

Next steps. We encourage future researchers to investigate whether mechanistic interpretability could yield answers to questions raised herein. It would be valuable to determine how models are currently performing long-context narrative analysis and tracking narrative state, and whether their representational mechanisms are similar to those of humans. Doing so would help researchers develop strategies for better predicting which long-context tasks LLMs are most appropriate for.

Limitations

There are several key limitations to this work. The first is the lack of true ground truth values. The expense and time needed to produce validated human ground truth for full novel-level annotations means that the TLDM benchmark compares novels only to their own short context performance; thus, while we are able to evaluate how model performance extends comparatively to long contexts, we lack maximally robust assessments of model vs. human capabilities. Second, all texts included in the benchmark are in English, meaning we do not evaluate models' multilingual performance. Finally, compute restrictions limit the number and variety of models we are able to evaluate in this paper.

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A Corpus Contents

The books included in the TLDM benchmark are:

<32k

- *Beasley’s Christmas Party* by Booth Tarkington
- *The Battle Of The Strong (A Romance of Two Kingdoms): Volume 2* by Gilbert Parker
- *The Caxtons: Part 12* by Edward Bulwer-Lytton
- *Godolphin: Volume 5* by Edward Bulwer-Lytton
- *The Romance of a Christmas Card* by Kate Douglas Wiggin
- *The Story of a China Cat* by Laura Lee Hope
- *Better Dead* by J. M. Barrie
- *The Tale of Jasper Jay Tuck-Me-In Tales* by Arthur Scott Bailey
- *Our Little Hawaiian Cousin* by Mary Hazelton Wade
- *Christmas at Thompson Hall* by Anthony Trollope

32k–64k

- *Alexander’s Bridge and The Barrel Organ* by Willa Cather and Alfred Noyes
- *Tom Swift and His Undersea Search or The Treasure on the Floor of the Atlantic* by